Engine Model(s): PT6A

74-10-00

# IGNITION EXCITER - MAINTENANCE PRACTICES

## 1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00, STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire used shall comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03, which is 0.025 inch diameter, and will not be specified in instructions.

## 2. Consumable Materials

The consumable materials listed below are used in the following procedures.

| Item No. | Name |
|---|---|
| PWC03-001 | Engine Oil |
| PWC11-012 | Acetone |
| PWC11-014 | Isopropyl Alcohol |
| PWC11-023 | Solvent, Cleaning |
| PWC11-025 | Solvent, Cleaning |

## 3. Special Tools

Not Applicable

## 4. Fixtures, Equipment and Supplier Tools

Not Applicable

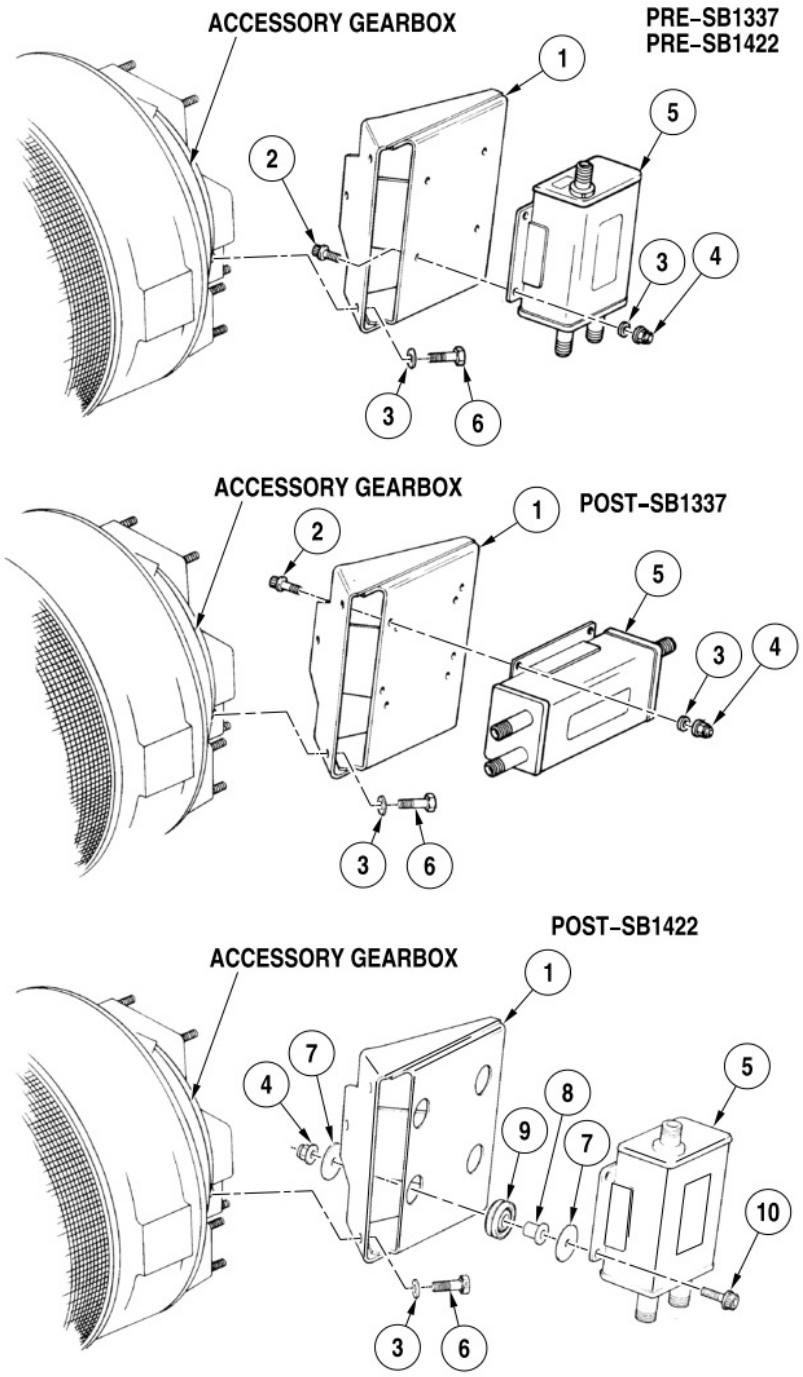
## 5. Removal/Installation

**WARNING: RESIDUAL VOLTAGE IN THE IGNITION EXCITER MAY BE DANGEROUSLY HIGH. MAKE SURE THE IGNITION IS SWITCHED OFF. ALWAYS DISCONNECT THE COUPLING NUTS AT THE IGNITION EXCITER END FIRST. ALWAYS USE INSULATED TOOLS TO REMOVE THE CABLE COUPLING NUTS. DO NOT TOUCH THE OUTPUT CONNECTORS OR COUPLING NUTS WITH BARE HANDS.**

A. Removal of Ignition Exciter (Ref. Fig. 201)


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**Figure 201 Removal/Installation of Ignition Exciter**




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1. Bracket Assembly
2. Bolt
3. Washer
4. Nut, Self-locking
5. Ignition Exciter
6. Bolt, Bracket Mounting
7. Washer (Post-SB1422)
8. Mount, Vibration Absorber (Post-SB1422)
9. Sleeve, Flanged (Post-SB1422)
10. Bolt (Post-SB1422)


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(1) Isolate the power from the ignition system.

(2) Remove the supply cable from the input connector on the ignition exciter (5).

**CAUTION: DO NOT ALLOW THE IGNITION CABLE BRAIDING OR FERRULES TO ROTATE WHEN RELEASING THE COUPLING NUTS.**

(3) Remove the two ignition cable couplings from the output connectors on the ignition exciter (5).

(4) Remove the four self-locking nuts (4), washers ((3) Pre-SB1422 or (7), Post-SB1422) and bolts ((2) Pre-SB1422 or (10) Post-SB1422), and remove the exciter unit (5) from the mounting bracket (1) on the accessory gearbox.

**NOTE**: The exciter may be remote-mounted on the airframe. Refer to the applicable Aircraft Maintenance Manual.

B. Installation of Ignition Exciter (Ref. Fig. 201)

(1) Install the ignition exciter on the mounting bracket on the accessory gearbox as follows:

(a) For Pre-SB1422 Engines: Install the ignition exciter mounting bolts (2) from the rear, through the bracket assembly (1) and ignition exciter (5) mounting flange. Secure with washers (3) and self-locking nuts (4), tighten 36 to 40 lb.in.

**NOTE**: Mounting bracket (1) (Post-SB1337) on PT6A-21/28 engines ( Beech Installation) has repositioned ignition exciter mounting holes.

(b) For Post-SB1422 Engines: Make sure the vibration absorption mounts (8) are secure in the bracket assembly (1). Insert the sleeves (9) into the vibration mounts and install the ignition exciter mounting bolts (10) through the ignition exciter (5) mounting flange, washer (7) and inserted sleeves in the vibration mounts. Secure with washers and self-locking nuts (4), tighten 36 to 40 lb.in.

**NOTE**: The exciter may be remote-mounted on the airframe. Refer to the applicable Aircraft Maintenance Manual.

**CAUTION: DO NOT LUBRICATE THE IGNITION EXCITER THREADS UNLESS THE TIN PLATING WAS REMOVED FROM THE THREADS (REF. PARA. 6.A.).**

**CAUTION: WHEN LOOSENING OR TIGHTENING THE COUPLING NUTS, DO NOT LET IGNITION CABLE BRAIDING OR FERRULES TURN AT SAME TIME.**

(2) Connect the coupling nuts of the supply cable and the two high tension ignition cable couplings to the respective input and output connectors on the ignition exciter. Tighten the nuts with your hand plus 45 degrees. Install safety wire.

**CAUTION: WHEN LOOSENING OR TIGHTENING THE COUPLING NUTS, DO NOT LET IGNITION CABLE BRAIDING OR FERRULES TURN AT SAME TIME.**

(3) Reconnect the coupling nuts at the other end of the ignition cables to the spark igniters. Tighten the nuts with your hand plus 45 degrees.


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## 6. Cleaning/Painting

A. Cleaning Ignition Exciter

(1) Remove all corrosion residue using a stainless steel wire brush.

(2) Clean the affected surfaces thoroughly with a clean lint-free cloth moist with acetone (PWC11-012), isopropyl alcohol (PWC11-014), solvent (PWC11-023) or (PWC11-025).

(3) Apply a thin layer of engine oil (PWC03-001) on the threads of the ignition exciter.

## 7. Inspection/Check

A. Inspection of Ignition Exciters

(1) Inspect the ignition exciters for signs of damage and general condition.

(2) Inspect the input and output connectors for damage, paying particular attention to the connector threads for corrosion.

B. Inspection of Ignition Exciter Attaching Parts

(1) Visually inspect bolts and nuts for condition of the threads. Replace as necessary.

(2) Visually inspect the bracket for cracks, distortion and general condition. Replace as necessary.

(3) Visually inspect the sleeves for cracks, distortion and general wear. Replace as necessary.

(4) Inspect the vibration absorption mounts for condition and resilience. Replace as necessary.


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